

Home Routines Suitable for Incidental Exposure to News: Pilot Field Experiment

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ABSTRACT

The explosion of media choices promotes "selective exposure;" entertainment-oriented people tend to consume only the information tied to their interests. It reduces their consumption and knowledge of broad info, including social issues. We consider putting information devices all around the house and proactively providing impartial news so that the residents could be incidentally exposed to broad knowledge while doing daily routines. A key to the success of the proactive presentation is providing information appropriately according to user contexts. Thus, we conducted field experiments with four participants to evaluate what user context suits incidental exposure. First, we implemented a tablet-based news app that continuously provides daily news from conspicuous locations in their homes. Experimental systems were installed in each participant's home where they spent about 10 days for the experiment. Before and after the experiment, participants answered questionnaires and participated in interviews. The results suggest features of the user context where the participants accept incidental exposure. Further, viewing content during daily routines could expand the news genres consumed by participants.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

KEYWORDS

Daily routines, News consumption behavior, Selective exposure, Incidental exposure

ACM Reference Format:

Hiromu Ogawa and Kinji Matsumura. 2022. Home Routines Suitable for Incidental Exposure to News: Pilot Field Experiment. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3544793.3560343>

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UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom
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ACM ISBN 978-1-4503-9423-9/22/09.
<https://doi.org/10.1145/3544793.3560343>

1 INTRODUCTION

Sharing social issues, a fundamental activity for democracy, is negatively impacted by "Selective exposure," the tendency of entertainment-oriented people to prioritize consuming entertainment content rather than political [1]. Current multiple media environments require users to voluntarily access news resource to get political information, but entertainment-oriented users are prone to avoid the access [1]. Less news consumption causes a decrease in political knowledge [1].

As a way to increase exposure to political news for even those disinterested in politics, incidental exposure has been studied. For example, Takano et al. show that even a short-duration exposure to political news during TV channel zapping enhances viewers' knowledge of political news [2]. However, the previous methods targeted only voluntary media use time. Thus, the opportunities for incidental exposure are limited to the time a user actively uses an information device.

To expand the chance of incidental exposure, we consider attaching information devices (i.e., tablets) at conspicuous locations in each room in a home (i.e., the washroom and kitchen) and proactively provide news. Thus, the residents can notice information while doing daily routines despite whether they intentionally watch them and consequently know the news. This method is expected to give the residents opportunities to see the news many times a day because people spend a lot of time on home routines; For example, Japanese adult women spend about 5 hours on housework on Sunday [3].

A key to the success of the proactive presentation is providing information appropriately according to user contexts; Notifications at inappropriate timing can be ignored by the user and reduce a user's satisfaction with the information system [4]. However, it is unclarified what daily routines suit the timing for incidental exposure. Further, its effect on interest in and knowledge of news is also unknown. Therefore, we conducted a field study to clarify them.

2 EXPERIMENT

The experiment was conducted in a home environment in which four tablets were separately installed in the washroom, kitchen, living room, and working space (1(a)). Each tablet ran an experimental news app that provided text-and image-based news continuously so that the participant could observe the news items while performing their normal daily activities. The app periodically crawled a news portal site and collected diverse news items, ranging in genre from politics to entertainment. The app consisted of two viewing modes: overview (Figure1(b-left)) and detailed (1(b-right)). The overview mode displayed news under 48-h old in 16 tile-shaped objects. Each



Figure 1: Experimental setting. (a) four tablets' placement; (b) news app appearance; (left) overview mode; (right) detailed view mode.

Table 1: Changes of participant interest in and knowledge of each news genre

ID	Living Information		Politics, Economics, Society		Hobby		Leisure		Sport		Entertainment		Trend		Local	
	I	K	I	K	I	K	I	K	I	K	I	K	I	K	I	K
P1	5(1)	4(1)	5(2)	4(2)	5(0)	4(0)	4(1)	3(0)	5(2)	4(2)	5(0)	4(0)	5(1)	4(1)	3(1)	2(0)
P2	5(0)	3(0)	5(0)	2(0)	5(0)	3(0)	5(0)	3(0)	1(-1)	1(0)	4(1)	2(0)	3(0)	3(0)	3(-1)	2(-1)
P3	5(0)	3(0)	5(1)	3(0)	5(1)	3(0)	5(1)	3(0)	4(2)	2(1)	5(0)	4(1)	5(1)	2(0)	5(0)	3(1)
P4	4(-1)	4(1)	4(1)	3(1)	5(0)	4(0)	5(0)	4(1)	1(0)	3(1)	2(0)	4(3)	4(0)	4(1)	5(0)	4(0)

tile contained the title and thumbnail of a news item, whose information was automatically swapped after being displayed for a random duration from 30 to 60 s or after its detailed view was selected by the participant. The detailed view appears after the viewer touches a tile.

Four participants participated in the experiment. They all lived in homes with two or three bedrooms with one or two family members. The experimental system was installed in each participant's house. The experimental duration ranged from 10 to 11 days per participant. During the experiments, participants were asked to continue their daily activities, and other family members were prohibited from operating the app. Informed consent was obtained from all participants regarding the collected data and academic usage purposes prior to the experiment.

Before and after the experiment, the participants rated their interest in and knowledge of eight news genres based on a scale of one to five. Additionally, after each experiment, we interviewed the participants to evaluate how news viewing during their daily activities affected their media consumption. Each semi-structured interview lasted about 70 min and was conducted by three experimenters (an interviewer, an assistant, and a recorder) with the single participant in Japanese.

3 RESULTS & DISCUSSION

Table 1 shows the results of the questionnaire and the changes in participants' interest in and knowledge of each news genre. The characters in the first column represent participant identifiers (i.e., "P1" stands for Participant 1). The first row represent news genres. The second row's characters represent evaluated items: "I" and "K" represent the participants' ratings of interest and knowledge, respectively. The other cell numbers (i.e., "5(1)") represent the ratings in the post-questionnaires (i.e., "5") and the amounts

of changes from the ratings in the pre-questionnaires (i.e., "(1)"). Each rating figure represents the following meanings; 1: "having no interest/knowledge," 2 "likely having no interest/knowledge," 3: "unsure," 4: "likely having interest/knowledge," and 5; "having interest/knowledge."

We describe only the summaries of typical questions and answers in interviews owing to space limitations.

We asked when they checked the news with the experimental system. All participants said they often checked provided information during routines in the washroom. Regarding the other three places, the opinion was divided; Some said they often checked the news there, and the others did not. According to their comments, acceptance of incidental exposure during routines depends on factors, including frequency to visit the place, the complication of the routine, and time of day.

Next, we asked how the incidental exposure affected the news genre consumed. P1 said that it induced him to check for broader news genres; He always brought his smartphone during staying at home. Sometimes he wanted to enjoy some content but could not afford to operate his phone and select content while doing daily routines. At the time, he naturally accepted the diverse news the system provided and consequently exposed diverse knowledge, including politics. P3 and P4 showed similar findings. Questionnaire results support that these three participants improved their interest in and knowledge of news from several genres. However, the system failed to expand P2's news consumption. She pointed out that she originally checked the news daily and was already knowledgeable; hence, the system failed to provide information that was unknown to her. Further analyses on the discrepancy are required.

4 CONCLUSION & FUTURE WORKS

The results of our pilot experiment indicate several features of home routines appropriate for content viewing. Further, it suggests incidental exposure during daily activities could induce the consumption of broad news, including politics. In the future, we will quantitatively evaluate the effect of content viewing on daily activities.

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